

Where Astra fails on engineering drawings

Problem statement, dataset, measured capability boundary and research focus

SVG Patch Lab • CVPR 2027 research package
branch `cvpr2027-research-package`
evidence in `runs/eng-frame-*` and `runs/eng-frame-aggregate.json`

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Status: a measured negative-and-positive result, not a submission. Every number is copied from the named run artefacts. Declining to answer and answering wrongly are reported as different outcomes throughout, because only the second puts a false claim on a drawing.

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1 Executive summary

On dimensioned structural elevations checked against a frame finite-element reference, GPT-6 Astra **declines to produce the required engineering quantities once the frame reaches about twenty members**, and does so on 4 of 7 samples at that size against 1 of 35 below it (Fisher exact, two-sided, $p = 0.0015$). It does not fabricate. Across 42 completed samples exactly one reported quantity fell outside tolerance, and that one did not reproduce on retest.

Three further findings shape how the result should be read.

1. **The two failures already recorded in the repository are not reproducible.** The headline 20 September result — a peak stress of 179 MPa against a reference of 20.356 MPa — came back correct on 3 of 3 fresh samples at the identical protocol. So did the first decline found in this study. Single-sample screening manufactures failures that evaporate.
2. **One apparent model failure was a defect in our own scorer.** Three results were rejected for omitting the unit mm on individual dimension labels, while each drawing declared its units once in the notes. That is standard drafting practice, and the harness was penalising correct work.
3. **Accuracy does not decay with complexity; willingness does.** When the model answers, it is within tolerance at every size tested, up to 21 members and degree of static indeterminacy 27.

The immediate consequence for the project is that the decline, not the wrong number, is the defect with a measurable baseline — and it is precisely the condition a solver-in-the-loop design is meant to remove.

2 Problem statement

2.1 The question

An engineering drawing is simultaneously a picture and a set of claims. A dimension is an assertion about a length; a reported stress is an assertion about a solved field. The claims can be false while the picture stays plausible, and editing the drawing — moving a grid line, raising a floor, changing a load — invalidates every derived claim at once without disturbing the geometry that carries them.

The programme’s question is whether a model can produce and edit such drawings so that their numerical claims remain true, and whether that truth can be *verified* rather than trusted. This study addresses the narrower empirical precondition:

On CAD-style engineering drawings whose claims are checkable by simulation, where exactly does the model stop being reliable, and what does failure look like when it arrives?

2.2 Why engineering drawings specifically

The artefact under study is a structural elevation of a physical object — a building frame, a worktable, a shelf — emitted as SVG and verified with finite elements. It is deliberately not a physics field visualisation. An earlier pass in this session built a benchmark on Laplace isotherm drawings, where a large and robustly reproducible failure does exist; that direction was abandoned because a contour plot is not an engineering drawing and the reproducible failure there answers a different question. Section 7.4 records it so the effort is not silently lost.

2.3 What counts as a failure

A drawing fails when a check that the task explicitly demanded does not hold: a member drawn away from its centreline, a joint that does not close, a dimension that disagrees with the geometry, a reported displacement or stress outside tolerance of the reference, or a visible SVG result that contradicts the returned JSON. Three outcomes are held apart, because the harness conflates them and the distinction carries the whole finding:

computed, within tolerance

every checked quantity agrees with the finite-element reference.

computed, outside tolerance

a number was printed on the drawing and it is wrong. This is the dangerous case: a released drawing carrying a false engineering claim.

declined

the model returned the contractually permitted `status:"unavailable"`, null values and a stated reason. The deliverable is incomplete, but no false claim was made.

Truncated responses, API errors and unparseable output are excluded from the denominators rather than counted as failures.

3 The task and how it is verified

3.1 What the model is asked to produce

Given a physical model in JSON — nodes, members with rectangular sections $b \times h$, clamped supports, nodal loads, a probe node — the model returns one JSON object containing a complete SVG and an analysis block. The drawing must place every physical member as a single visibly stroked straight element tagged `data-member`, retain member identifiers, honour a fixed `viewBox` and a fixed affine map from millimetres to drawing coordinates, and annotate the centreline length of every member as text tagged `data-dimension`. The analysis must report u_x and u_y at the probe node and the peak extreme-fibre normal stress

$$\sigma_{\max} = \max_{\text{member ends}} \left(\frac{|N|}{A} + \frac{|M| h/2}{I} \right),$$

together with a status of *calculated*, *estimated* or *unavailable*, and must repeat those three numbers as visible SVG text tagged `data-result`. No tools are supplied and the system prompt forbids claiming that a solver was run. In edit mode the model additionally receives the original model and its SVG, plus an instruction that changes geometry and a load, and must recompute every dimension and both analyses.

3.2 What is checked

Scoring reuses `cad_astra_benchmark.score` unchanged, so results are directly comparable with the 20 September pilot. It parses the SVG with explicit rejection of scripts, stylesheets, clipping, masks, `use` elements and hidden geometry; recovers each tagged member through its accumulated ancestor transform; and checks member endpoints and straightness to 1 mm, joint closure to 1 mm, every dimension to 1 mm, the three analysis quantities against the reference at 2% relative tolerance, and agreement between the JSON numbers and the visible SVG text to 0.1%.

The reference is a linear plane-frame finite-element solution — axial plus Euler–Bernoulli bending, rigid joints, small displacements, nodal loads — assembled independently of the model’s output.

4 Dataset: EngFrame

4.1 Design principle: order the family by static indeterminacy

The pilot contained three hand-built families and exactly one failure. Ordering them by degree of static indeterminacy for a plane frame,

$$i = 3m + r - 3j,$$

with m members, r reaction components and j joints, reproduces their outcomes exactly:

Pilot family	Members	i	Pilot outcome
shelf (cantilever chain)	4	0	pass
table (portal frame)	4	3	pass
building (two-storey setback)	8	9	the sole failure

The axis is mechanically meaningful rather than fitted after the fact: a determinate frame ($i = 0$) is solvable by statics alone, whereas an indeterminate one requires the simultaneous stiffness system, which is what must be carried out without a solver. EngFrame generates frames along that axis.

4.2 The generator

`scripts/eng_frame_bench.py` builds fixed-base rectilinear building frames from a small parameter set: number of bays, number of storeys, bay width, storey height, and an optional setback that removes the rightmost column lines above a chosen floor — the shape of the pilot’s two-storey building. Column and beam sections thin with height. Loads are applied at the outer nodes of every floor. For a plain rectangular fixed-base frame with n_b bays and n_s storeys the construction gives $i = 3n_bn_s$ exactly, so the difficulty axis is controlled by construction rather than measured after generation.

Each family is emitted twice, in *generate* and *edit* mode, where the edit shifts a column grid by 700 mm or raises the top floor by 600 mm and changes a nodal load by 15 kN. Verification requires that every edit changes both the dimensions and all three analysis quantities, so that no instance can be satisfied by copying the source.

4.3 Reference certification

A reference is admitted only if it survives two independent checks: member-subdivision invariance, requiring the nodal response with each member split into four elements to agree with the single-element assembly to a relative 10^{-7} ; and global force and moment equilibrium, requiring the residual to be below 10^{-9} of the largest applied load component. Eight further manifest checks pass, among them that every frame is well formed with its recorded indeterminacy correct, that every reference design satisfies its own stated limits, and — the positive control — that the harness scores a drawing constructed from the reference itself as a pass.

4.4 Composition

Instances are concentrated in the hard region on purpose. There is no value in re-measuring the region the model already solves, so 14 of the 18 main instances sit at $i \geq 9$, with a separate extension set at $i = 24$ and 27. The extension set deliberately carries no control; its manifest records that controls live in the companion main set, and verification enforces that pairing rather than waiving it.

Family	Members	Nodes	Supports	i	Reference $\sigma_{\max}$ (MPa)
<i>Main set — 9 families, 18 tasks</i>					
control_portal	3	4	2	3	9.550
control_twobay	5	6	3	6	6.233
setback_a	8	8	3	9	14.934
setback_b	8	8	3	9	15.829
threestorey	9	8	2	9	33.540
twostorey_a	10	9	3	12	13.863
twostorey_b	10	9	3	12	13.215
setback_tall	13	12	4	15	31.616
threebay	14	12	4	18	10.860
<i>Extension set — 3 families, 6 tasks</i>					
fourbay	18	15	5	24	9.110
twobay_four	20	15	3	24	33.316
threebay_three	21	16	4	27	17.185

5 What Astra cannot do

5.1 The boundary

Forty-five samples were collected, of which 42 completed; the remaining three were cut off by credit exhaustion and are excluded. Outcomes group sharply by frame size.

Members	Samples	Correct	Outside tolerance	Declined	Decline rate
3–13	25	25	0	0	0 %
14	8	7	0	1	12 %
18	2	2	0	0	0 %
20	2	1	0	1	50 %
21	5	2	0	3	60 %
< 20	35	34	1	1	3 %
≥ 20	7	3	0	4	57 %

The contrast between the two pooled rows is significant at $p = 0.0015$ by a two-sided Fisher exact test. The decline rate above the boundary is 57 % with a 95 % Wilson interval of 25–84 %.

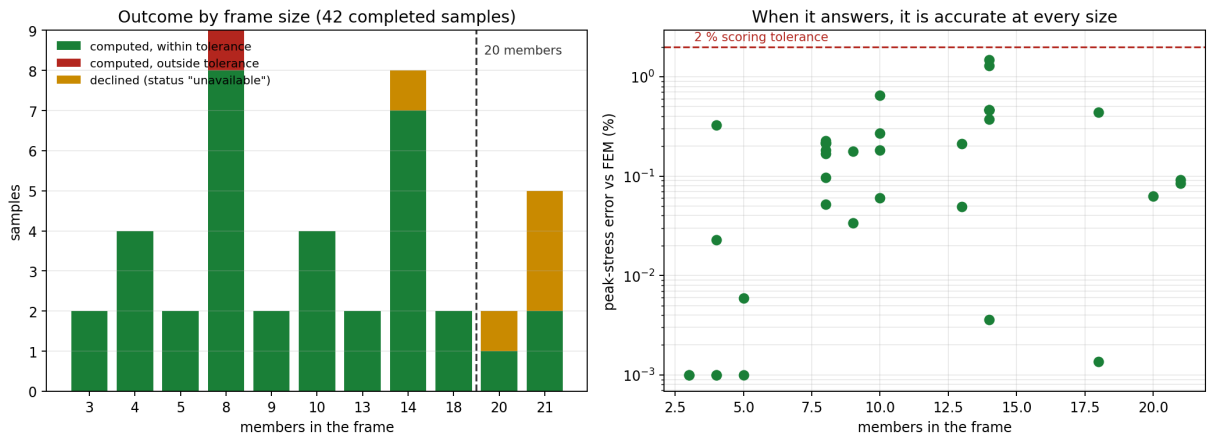


Figure 1: Left: outcome by frame size across 42 completed samples; declines appear only at the top of the range. Right: for every sample where a number was given, its peak-stress error against the finite-element reference. All 36 lie below the 2 % tolerance and show no trend with size.

5.2 What the failure looks like

The declines are explicit and honest. The model sets the permitted status, returns nulls, and states a specific reason — for the 21-member edit, “*the edited frame stiffness solution and member-end force recovery have not been performed*”; for the 20-member generation, “*the coupled axial-bending frame response has not been computed*”. It identifies precisely the step it did not carry out. Token budget is not the constraint: declines finish cleanly with `finish_reason=stop` after 6,600–8,100 completion tokens against a 32,000 cap, while successful answers on the same instances use 9,000–13,000.

This is a capability boundary in *willingness to carry out an unaided indeterminate solve*, not in drafting. In every declining sample the geometry and the dimensions were still produced.

5.3 What it can do, for scope

The claim must be bounded by an equally clear statement of competence. Across this study the model drew every frame correctly, at every size, in both modes. When it reported numbers they were within tolerance in all 36 cases, including at 21 members and $i = 27$, where it returned a peak stress of 17.7 MPa against a reference of 17.6837 and u_x of 3.5 against 3.5119. On the pilot’s failing instance it had already identified the correct governing member end and stated the correct extreme-fibre formula. Prior work in this package further records that the model solves PCB routing with clearance and exact-length constraints, irregular part nesting, multiview machining edits and FEM-constrained section sizing. The honest summary is that this task family is close to solved, with one measurable exception.

6 Negative results and two corrections

6.1 Recorded failures that did not reproduce

Lead	Recorded result	On retest at identical protocol
building_edit, 20 Sep pilot	peak stress 179 MPa against a reference of 20.356 MPa	3 of 3 samples correct (20.4 MPa)
threebay_edit, this study	analysis declined	3 of 3 samples correct

The 179 MPa case remains the most diagnostic artefact in the repository. Geometry and dimensions were correct, both displacements were right to 0.3%, the governing member end was correctly named, and the formula was correctly stated — and the number printed was 8.8 times too large. It is a genuine defect and it is not reproducible, so under the pilot’s own admission rule it is not an Astra-hard case and must not be cited as one.

6.2 A harness defect mistaken for a model defect

Three screened results were filed as unscored format failures for “missing mm unit” on every dimension label. Inspection shows each drawing declares its units once, for instance “*Structural elevation · dimensions in mm · member centrelines*”. Stating units once in the notes rather than repeating them on every dimension is standard ISO and ASME practice, so the harness was rejecting correct drafting. `eng_frame_bench.adjudicate` withdraws that complaint, but only when the drawing declares its units and nothing else fails; it can never convert a failure into a pass on any other ground. One of those three, `twobay_four_generate`, proved to have declined its analysis as well — the unit complaint had masked the real defect.

6.3 Methodological consequence

Three of the first four anomalies found in this study disappeared on repetition. Any screen reporting one sample per task will therefore report failures that do not exist, and — since a 57% decline rate still yields a pass 43% of the time — will also miss real ones. Confirmation at three or more fresh samples is not bureaucratic caution here; it is the difference between the two headline leads being real and being noise. Both were noise.

7 Our focus

7.1 The immediate experiment

The decline is the right target, for three reasons. It is statistically established rather than anecdotal; it is the only measured boundary in a family the model otherwise solves; and it is exactly the condition that the project’s solver-in-the-loop thesis predicts should be removable. The experiment is direct: re-run the instances at and above 20 members with a numerical tool available — a stiffness solve or member-end force recovery the model may call — and measure whether the decline rate falls while accuracy is preserved. A baseline of 4 declines in 7 samples now exists to measure against.

Two secondary arms follow cheaply: raising the reasoning effort and the token cap at fixed tool access, to separate unwillingness from budget; and extending the ladder beyond 21 members to locate the threshold, which the present data brackets only loosely between 18 and 20.

7.2 Why this reframes the project’s premise

The programme was organised around the hypothesis that the model would draw plausibly and be *numerically wrong*, so that binding claims to a verified solution would catch silent errors. The measurement does not support that in this family. The model is numerically right whenever it commits, and when it cannot commit it says so. The valuable contract is therefore not primarily a lie detector but a *completion* mechanism: the verification layer supplies the solve the model declines to attempt, and the drawing is finished rather than rejected. That is a narrower and better-evidenced claim than the original framing, and it is the one the artefacts here support.

The one recorded instance of a confidently wrong number, at 1 in 42 samples, keeps the verification argument alive but cannot carry it alone; establishing that rate would need roughly an order of magnitude more samples than this study could fund.

7.3 Relation to existing work

Combining a language model, a simulator and vector graphics is not novel, and the package’s prior-art review already records close precedents: EngDesign for functionally simulated design tasks, StructureClaw for linked structural artefacts with numerical checks, ALL-FEM for fine-tuned solver-code agents, PDEAgent-Bench for calibrated PDE references, and SSVG-Bench for structured scientific vector graphics. What this study contributes is narrower and empirical: a procedurally generated engineering drawing family ordered by a mechanically meaningful difficulty parameter, with certified references, and a measured capability boundary with its statistics and its non-reproducing counterexamples stated. That is a benchmark and a measurement, not a method claim.

7.4 The abandoned direction

A benchmark on Laplace isotherm drawings over procedurally generated rectilinear plates was built and verified before being set aside as off-target. It is retained because its findings are

independently useful: the generator reproduces the 18 September failing task character for character; the worst grid sensitivity of the finite-difference reference sits exactly on the re-entrant corners, which therefore must be excluded from any scored metric; and the strict criterion used on 18 September is unreachable on that domain, since a drawing built from the reference isolines themselves scores 5.0003 percentage points against a 0.5 point tolerance. The 90.0 point maxima reported there are that artefact, not a finding. Artefacts are in `data/polyheat-bench/` with 15 passing tests.

8 Limits

Seven completed samples above the boundary is a small denominator, and the 25–84% interval is correspondingly wide. The threshold is bracketed, not resolved: 18 members passed twice, 20 and 21 declined in 4 of 7 samples, so “about twenty members” is the defensible statement and the confound between member count and degree of indeterminacy is not separated by these data. The `twobay_four_generate` confirmation was cut off by credit exhaustion, leaving that instance with a single screened sample; its three attempts are recorded as excluded API errors and the runner will not re-bill them, so completing it requires a fresh output directory.

All runs used `gpt-6-astra` at medium reasoning effort through Chat Completions with `store=false`. Whether a larger budget, higher effort or tool access removes the decline is untested. The structural model is linear, planar, rigid-jointed and small-displacement with nodal loads and rectangular sections; self-weight, shear deformation, buckling, out-of-plane behaviour, connection flexibility and code compliance are all outside it. The scoring is a geometric and numerical audit of tagged elements, not a rendered-image review: occlusion, legibility, line weights, layer conventions and title-block practice are not assessed. Nothing here bounds performance on 3D CAD or on drawing families other than planar frame elevations.

9 Artefacts and reproduction

Path	Contents
<code>scripts/eng_frame_bench.py</code>	generator, reference certification, 8 verification checks, units adjudication
<code>scripts/eng_frame_report.py</code>	cross-run aggregation separating declined from wrong
<code>data/eng-frame-bench/</code>	main set, 18 tasks, $i = 3-18$
<code>data/eng-frame-bench-xl/</code>	extension set, 6 tasks, $i = 24-27$
<code>runs/eng-frame-*/</code>	screens, confirmations, per-sample requests and responses
<code>runs/eng-frame-aggregate.json</code>	the 45-sample table behind every figure here
<code>reports/ENG_FRAME_FAILURE_BOUN</code>	the prose record
<code>DARY_20260922.md</code>	

```
cd cvpr2027
PYTHONPATH=src ../.venv/bin/python scripts/eng_frame_bench.py build
PYTHONPATH=src ../.venv/bin/python scripts/eng_frame_bench.py build --set xl \
  --data data/eng-frame-bench-xl
PYTHONPATH=src ../.venv/bin/python scripts/eng_frame_bench.py verify
PYTHONPATH=src ../.venv/bin/python scripts/cad_astra_benchmark.py run \
  --data data/eng-frame-bench --output runs/<fresh> --samples 3      # billable
PYTHONPATH=src ../.venv/bin/python scripts/cad_astra_benchmark.py score \
  --data data/eng-frame-bench --output runs/<fresh>
```

```
PYTHONPATH=src ../.venv/bin/python scripts/eng_frame_bench.py adjudicate --output  
↪ runs/<fresh>  
PYTHONPATH=src ../.venv/bin/python scripts/eng_frame_report.py
```

build is deterministic. run resumes without re-billing a sample whose result exists, and refuses a directory created under a different protocol.

10 Provenance and cost

This study used 115,455 input and 375,052 completion tokens, approximately **\$19.91** at the documented list prices of \$10 and \$50 per million tokens. That is token-price arithmetic, not an invoice or a balance reading. Of it, \$1.18 went to the abandoned isotherm direction and \$0.87 to a void run whose prompts omitted the `data-level` contract and which is retained, clearly marked, at `runs/VOID-polyheat-zeroshot-20260922-missing-status-line/`. The account returned `credit_balance_exhausted` during the final confirmation, which is why one instance remains incomplete.

All model outputs, requests, protocols and scores are retained unmodified. Where a verdict was changed after the fact — only the withdrawal of per-label unit complaints — the original score, the new score and the reason are stored side by side in the run's `adjudication.json`.